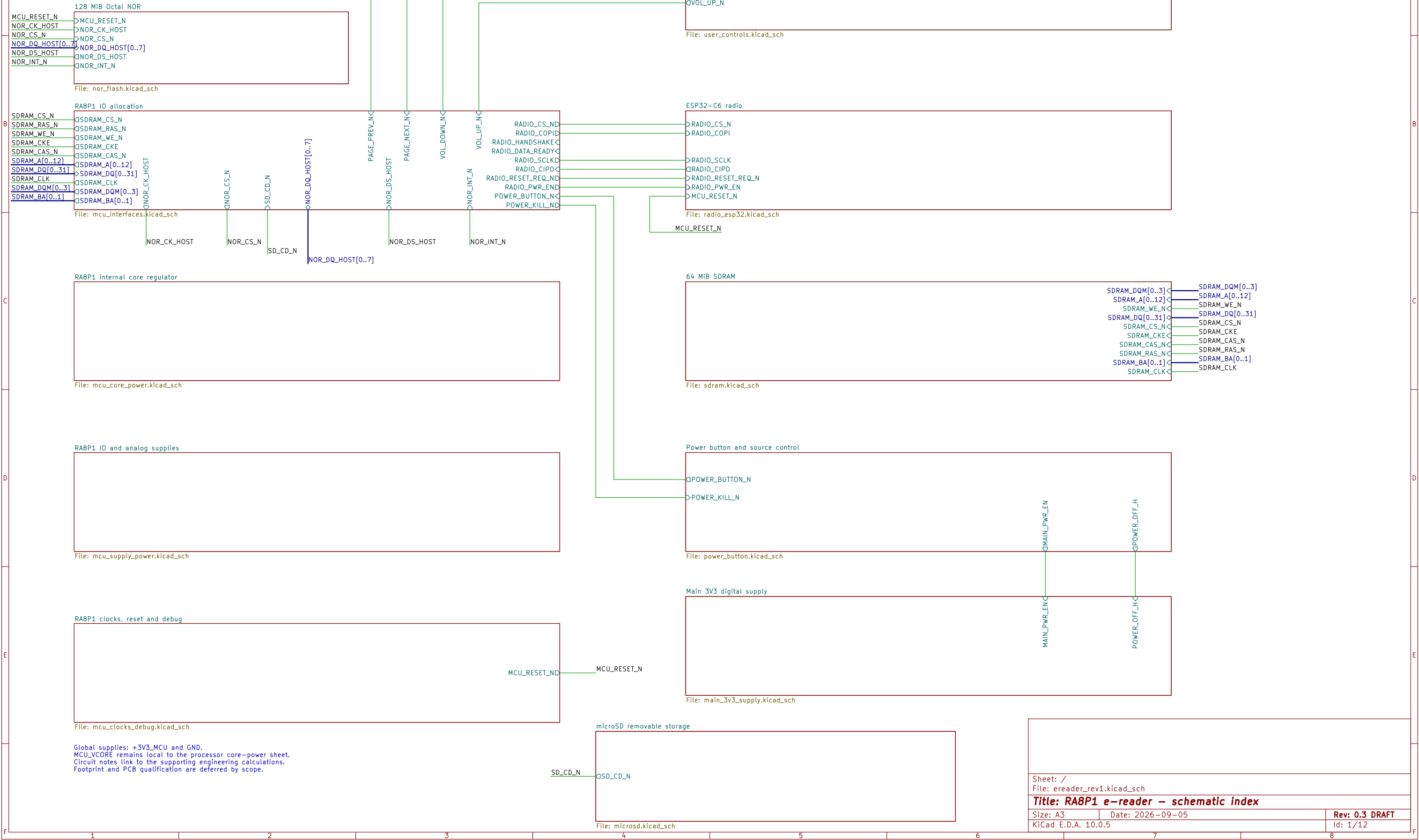
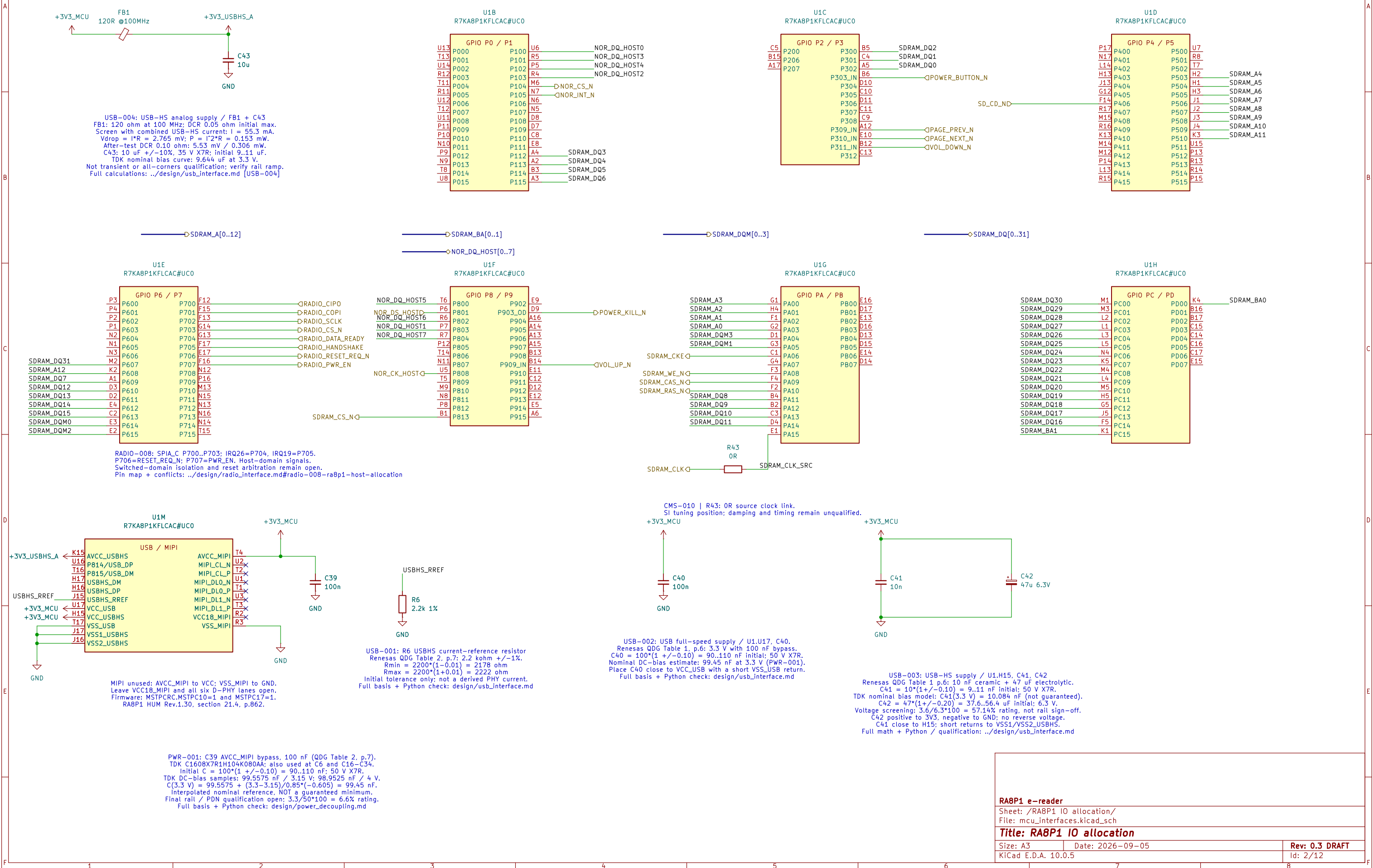


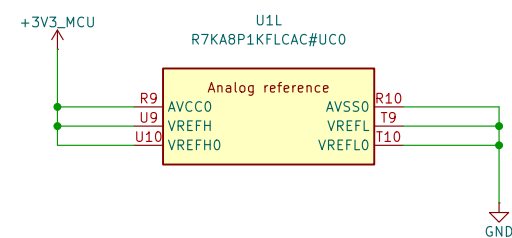
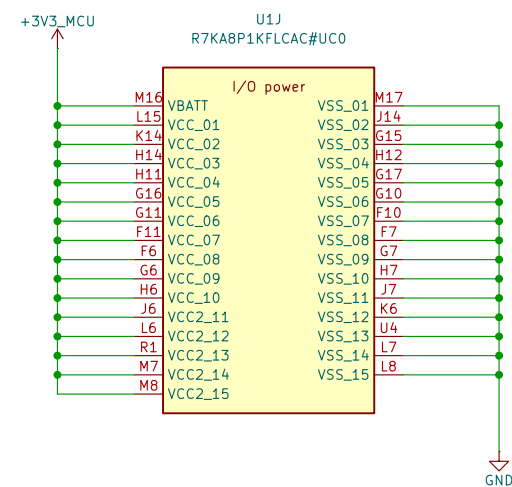
RA8P1 E-READER

Development draft – not for fabrication.
Requirements: design/ereader_requirements.md
Tracking: GitHub epic #821.

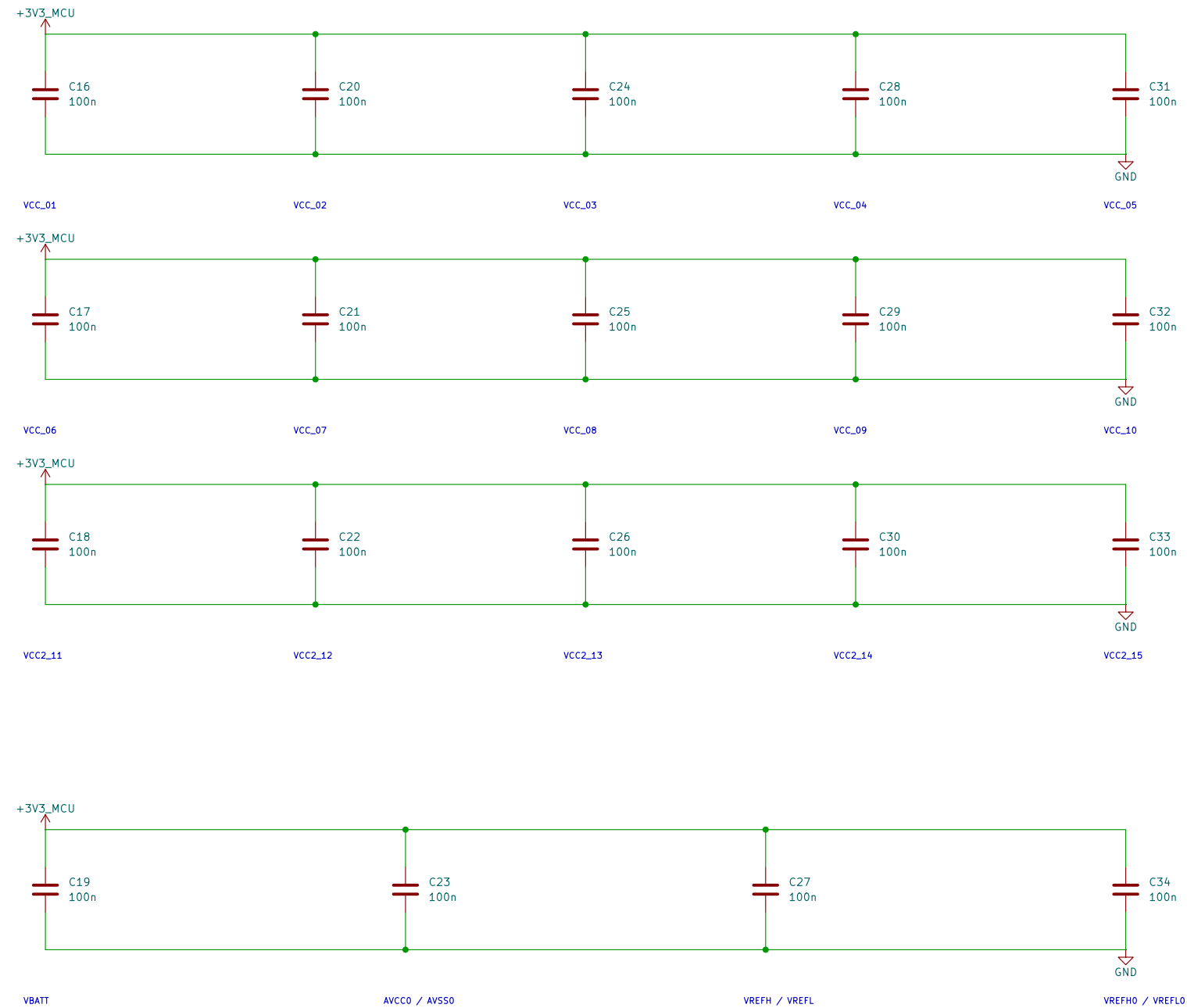


DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

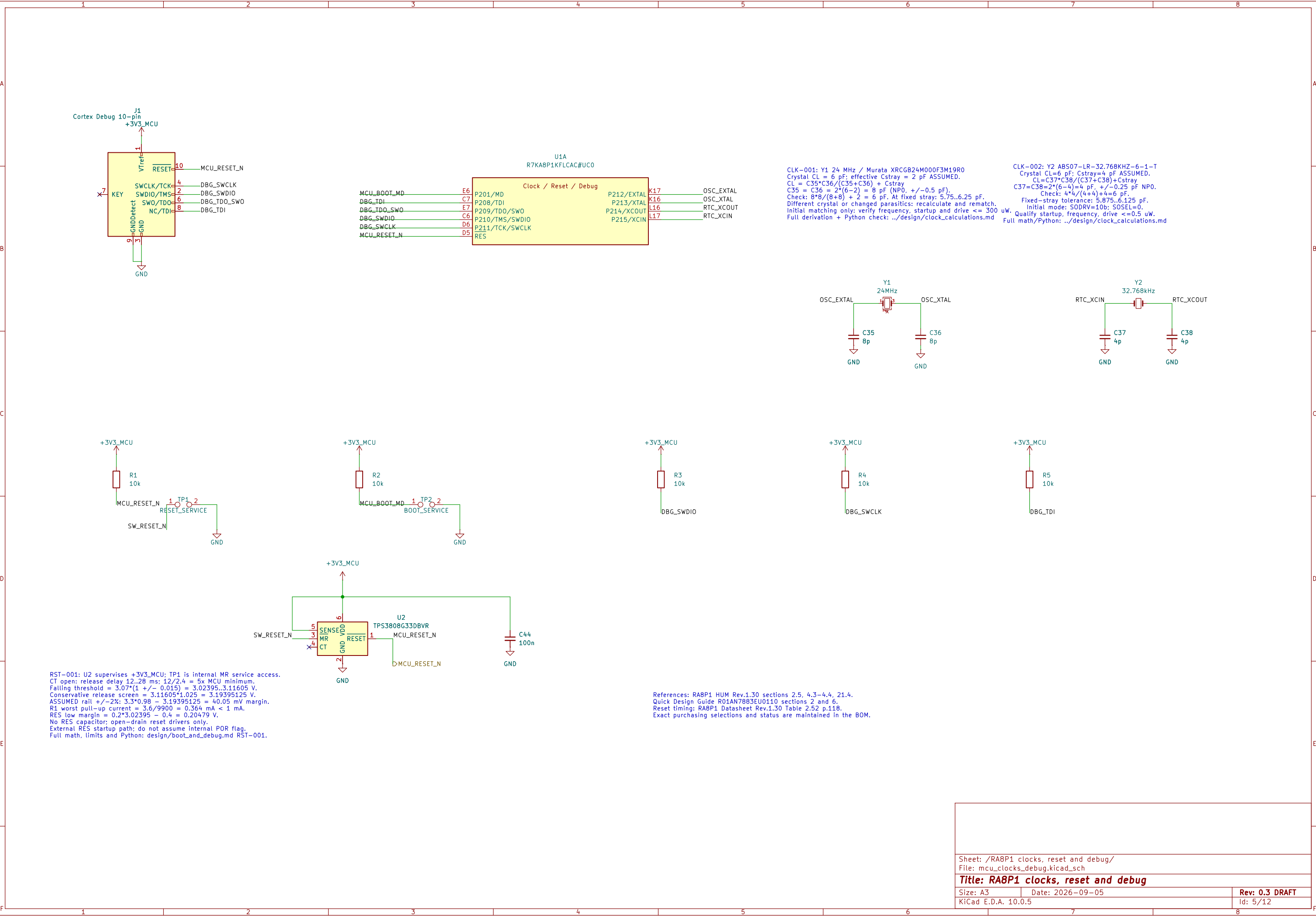


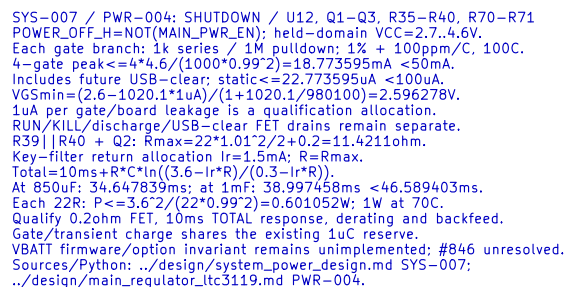
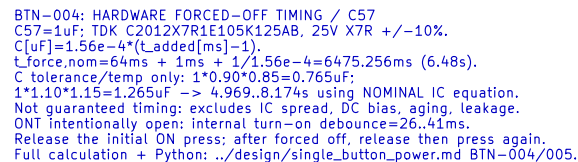


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

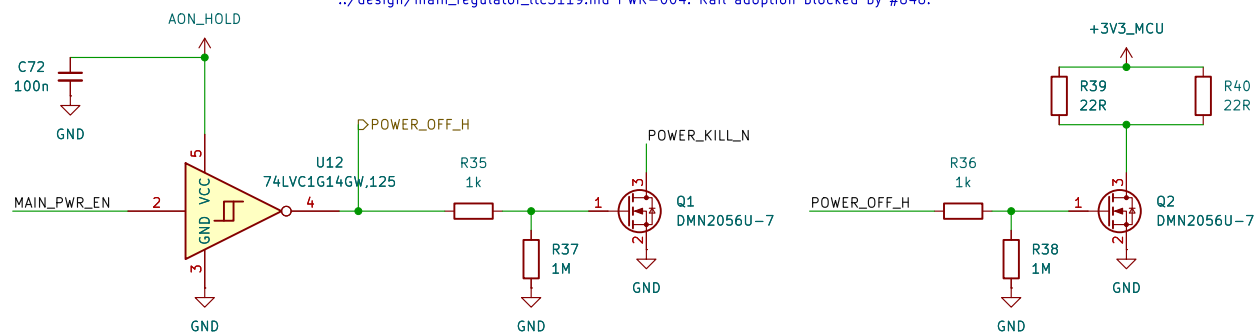
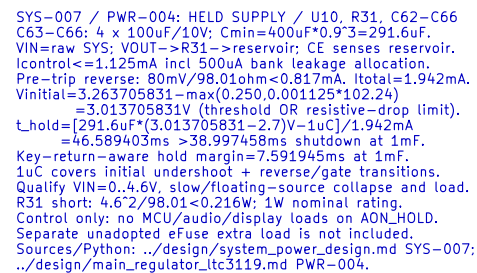


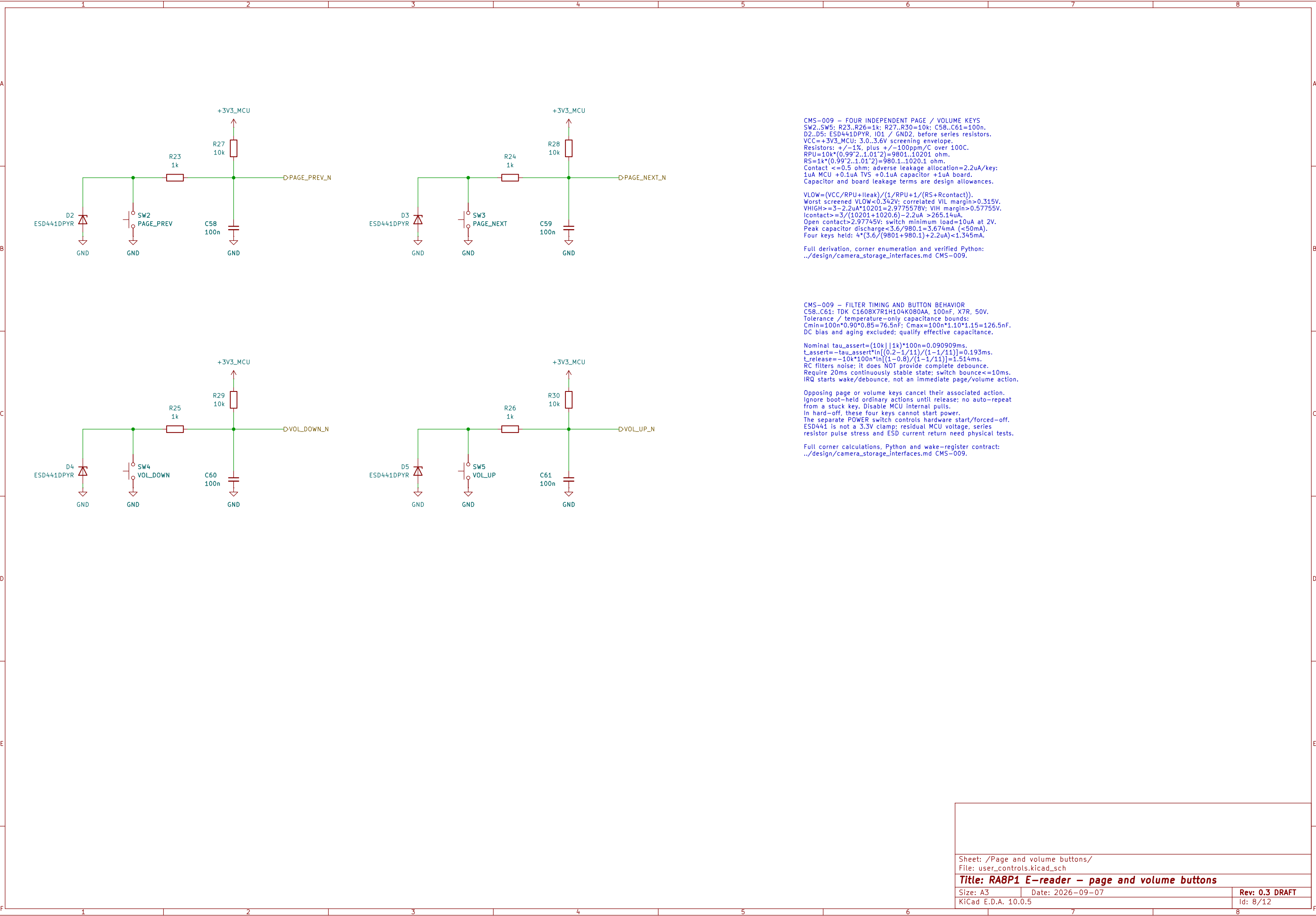
VCC and VCC2 use 3.3 V. All returns use one board ground plane.
VBATT follows the main rail; no separate backup-cell supply.
Analog references use AVCC0 / AVSS0. Place bypasses at named ball pairs.
USB/MIP1 supplies remain on the interface sheet and are not yet wired.
HUM Rev.1.30: Table 69.2 p.4042; unused references: section 21.4 pp.861–862.
This is an electrical draft; capacitor MPN qualification remains open.





ERC supply path: U10 VOUT through R31.
PWR_FLAG declares that series-fed source only;
raw SYS_AON source remains unimplemented.





CMS-009 – FOUR INDEPENDENT PAGE / VOLUME KEYS
SW2..SW5; R23..R26=1k; R27..R30=10k; C58..C61=100n.
D2..D5: ESD441DPYR, IO1 / GND2, before series resistors.
VCC=+3V3_MCU; 3.0..3.6V screening envelope.
Resistors: +/-1%, plus +/-100ppm/C over 100C.
RPU=10k*(0.99^2..1.01^2)=9801..10201 ohm.
RS=1k*(0.99^2..1.01^2)=9801..10201 ohm.
Contact <=0.5 ohm; adverse leakage allocation=2.2uA/key:
1uA MCU +0.1uA TVS +0.1uA capacitor +1uA board.
Capacitor and board leakage terms are design allowances.

$VLOW=(VCC/RPU+Ileak)/(1/RPU+1/(RS+Rcontact))$.
Worst screened VLOW<0.342V; correlated VIL margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; VIH margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact>2.97745V; switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/(9801+980.1)+2.2uA)<1.345mA.

Full derivation, corner enumeration and verified Python:
../design/camera_storage_interfaces.md CMS-009.

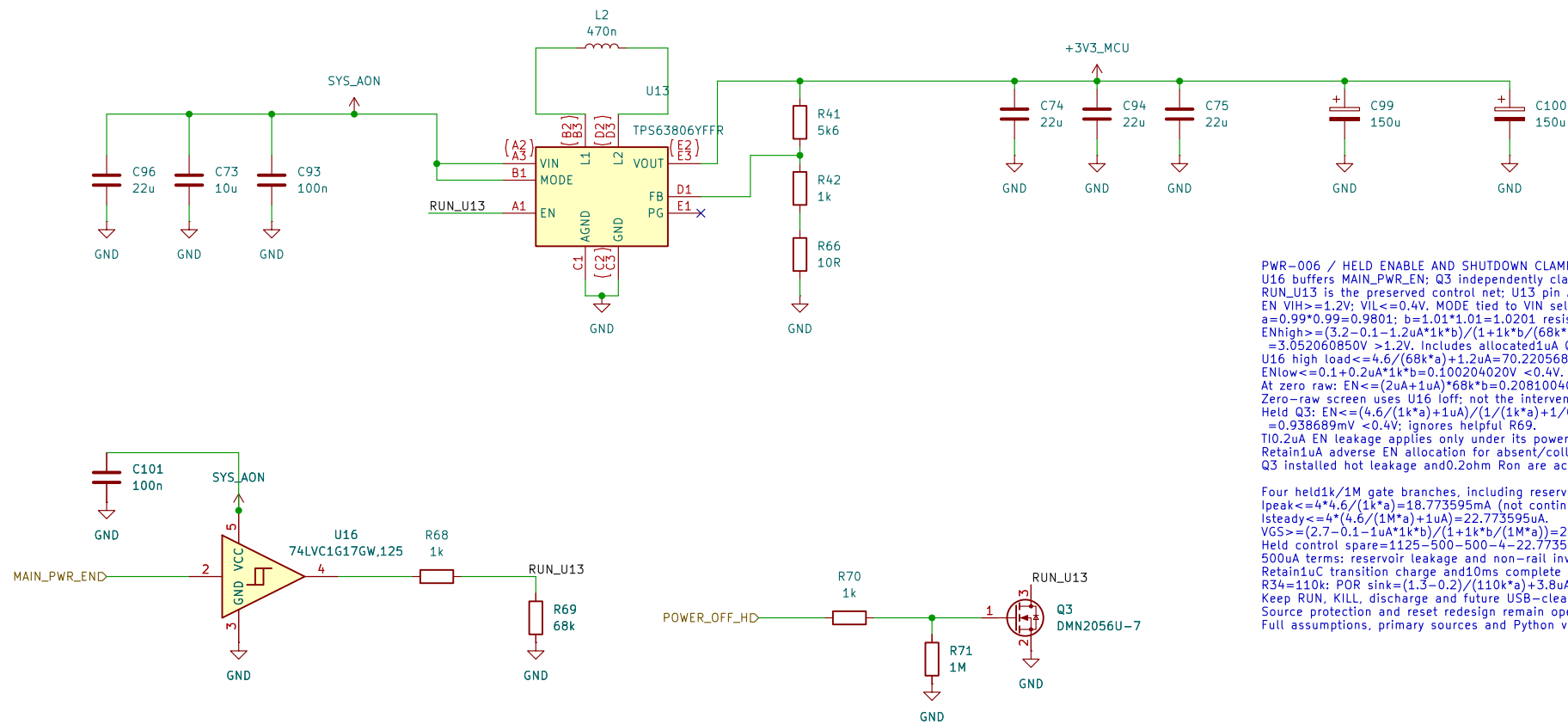
CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
Cmin=100n*0.90*0.85=76.5nF; Cmax=100n*1.10*1.15=126.5nF.
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k||1k)*100n=0.090909ms.
t_assert=-tau_assert*ln[(0.2-1/11)/(1-1/11)]=0.193ms.
t_release=-10k*100n*ln[(1-0.8)/(1-1/11)]=1.514ms.
RC filters noise; it does NOT provide complete debounce.
Require 20ms continuously stable state; switch bounce<=10ms.
IRQ starts wake/debounce, not an immediate page/volume action.

Opposing page or volume keys cancel their associated action.
Ignore boot-held ordinary actions until release; no auto-repeat
from a stuck key. Disable MCU internal pulls.
In hard-off, these four keys cannot start power.
The separate POWER switch controls hardware start/forced-off.
ESD441 is not a 3.3V clamp; residual MCU voltage, series
resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN.PWR.EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=+1.2V; VIL<=0.4V. MODE tied to V1N selects forced PWM.
a=0.99*0.99+0.9801; b=1.01*1.01-1.0201 resistor factors.
ENhigh>=(3.2-0.1-1.2uA*1k*b)/(1+1k*b*(68k*a))
=3.052060850V >1.2V. Includes allocated uA Q3 off leakage.
U16 high load <=4.6/(68k*a)+1.2uA=70.220568uA <100uA.
ENlow<=0.1+0.2uA*1k*b+0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*a)+1uA)/(1/(1k*a)+1/0.2ohm)
=0.938689mV <0.4V; ignores helpful R69.
T10.2uA EN leakage applies only under its powered test conditions.
Retain1uA adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and0.2ohm Rn are acceptance allocations.

```

```

Four 4h1dk/1M gate branches, including reserved USB-clear:
Ipeak<=4*4.6/(1k*1a)=18.773595mA (not continuous rating).
Isteady<=4*4.6/(1M*1a)+1A=22.773595mA.
VGS>=(2.7-0.1-1uA/1k*1b)/(1+1k*b/(1M)*a)=2.596278V.
Held control spare=1125-500-500-4-22.773595=98.226405uA.
500uA terms: reservoir leakage and non-rail inverter-injection allocation.
R34=10k: transition resistance and gate leakage.
R34=110k: POR sink=(1.3-0.2)/(110k*1a)+3.8uA=14.003041uA.
Keep RUN, KILL, discharge and future USB-clear drains separate.
Source protection and reset redesign remain open; no power-up approval.
Full assumptions, primary sources and Python verification: click this note.

```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR100C" aging*assembly, then ohms.
 $R41 = 5600 \times (1 + / - 0.0002) \times (1 + / - 0.0005 \times (1 + / - 0.001))$
 $\times (1 + / - 0.0005) + / - 0.02 = 5587.669237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + / - 0.0001) \times (1 + / - 0.0002) \times (1 + / - 0.001)$
 $\times (1 + / - 0.0005) + / - 0.01 = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + / - 0.01) \times (1 + / - 0.02) \times (1 + / - 0.001) + / - 0.05$
 $= 55.55555556, 55.55555556 \text{ ohm}$
V60: 0.01%, 1ppm/C; V45020: 0.01%, combined aging/assembly 1%.
 $V60 = 0.01 \times (1 + 141 \text{ min} / (R42 \text{ max} + R66 \text{ max})) \times 100 \text{ ohm} \times \text{max}$
 $V60 = 5.005 \times (1 + R41 \text{ max} / (R42 \text{ min} + R66 \text{ min})) + 100 \text{ nA} \times R41 \text{ max}$
Vstatic = 3.226819680, 3.318012496 V (FPWM only).
One combined + / - 75mV regulation/ripple/transient/routing allowance.
Vcomplete = 3.151819680...3.393012496 V: qualification required.
SDRAM static VOH margin = 2.4 - 0.73 = 3.93012496 = 2.4891253mV.
This is not measured S/I/noise overhead or reset qualification.

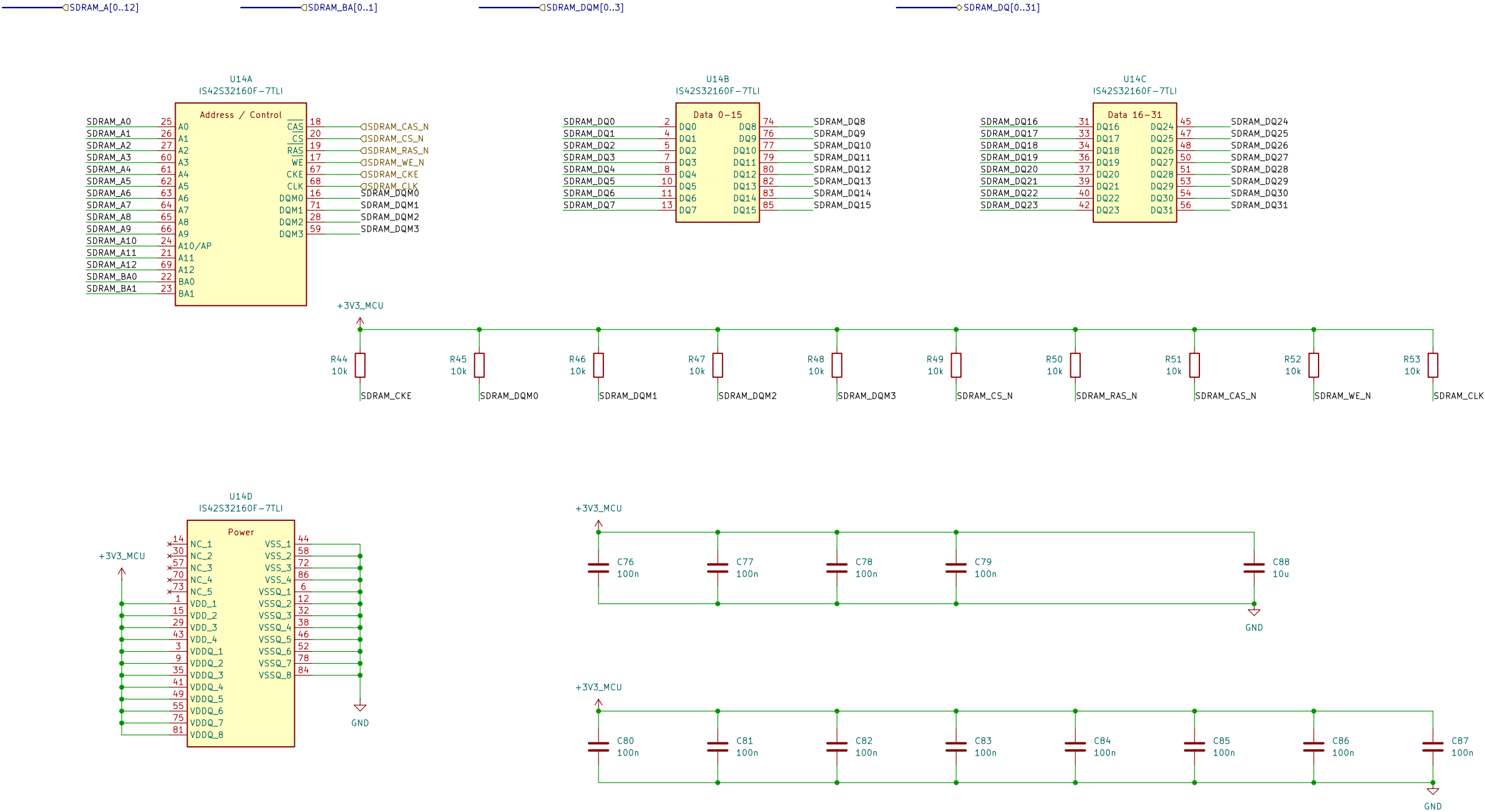
12. 470nH +/-20%; catalog spread alone is insufficient.
 Screen incoming/0.412-0.518uH; allocated operating +/-10%
 =>0.3708-0.5698uH versus 0.1837-0.57uH. Quality hot/current.
 C74/C75/C94: $3.22uF \pm 0.85\% = 6.2928uF \pm 121uF$.
 C96: $22uF \pm 0.85\% = 6.8796uF$. T4uF input minimum.
 C97: 100nF combined input-output aging retention=100nF.
 Samsung typical output-bias retention at3.393012496V=90.221554%.
 Input bias, aging, mounted impedance and startup need verification.
 Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.8A operating; 2.25A cold start screen, not measured load.
Pout=3.393012496*1.8=6.107422493W
At allocated 75% efficiency: Pin=8.14329991W; lin@3.2V=2.544759372A.
Loss=2.035807498W; NOT an acceptable continuous thermal result.
At 90% efficiency: loss=0.678602499W.
TJ screen=70C+78.8C/W*0.678602499W=123.474C.
TJ board theta-JA is not the final PCB thermal resistance.
2.25A screen: Pin=3.393012496*2.25/0.75=10.179037489W.
Converter limit, source limit, startup and thermal envelope need validation.

Native storage inventory (nominal, not effective minimum):
 Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
 =479.61uF cold; radio -on adds0.2uF =>489.81uF
 C99/C100 remain150uF each. C94 adds22uF to the output.
 Old bootstrap/private VCC/VIC capacitors were not main-rail storage.
 Whole-main ceiling remains1mF for shutdown; no LTC stability transfer.
 Cout lower acceptance and distributed PDN/startup are separate checks.

Preserved discharge screen: Rmax1,4211mohm, qny return<=1.5mA, Vinf=1.5mA*1,4211mohm=0.0713165V.
 t=mpf*1,4211mohm*ln((3.6-Vinf)/(0.3-Vinf))+10ms
 =39.97ms8ms, to 0.3V, before head=559403ms screen.
 t=1msf14211 includes command <= clamp/discharge delays.
 Add budget retainsuic transition charge and98.226405uA spare.
 SYS_AON source protection, loaded startup and fast faults remain open.
 The J2/J3/4/5/6/7/8/9/10/11/12/13/14/15/16/17/18/19/20/21/22/23/24/25/26/27/28/29/30/31/32/33/34/35/36/37/38/39/40/41/42/43/44/45/46/47/48/49/50/51/52/53/54/55/56/57/58/59/60/61/62/63/64/65/66/67/68/69/70/71/72/73/74/75/76/77/78/79/80/81/82/83/84/85/86/87/88/89/90/91/92/93/94/95/96/97/98/99/100/101/102/103/104/105/106/107/108/109/110/111/112/113/114/115/116/117/118/119/120/121/122/123/124/125/126/127/128/129/130/131/132/133/134/135/136/137/138/139/140/141/142/143/144/145/146/147/148/149/150/151/152/153/154/155/156/157/158/159/160/161/162/163/164/165/166/167/168/169/170/171/172/173/174/175/176/177/178/179/180/181/182/183/184/185/186/187/188/189/190/191/192/193/194/195/196/197/198/199/200/201/202/203/204/205/206/207/208/209/210/211/212/213/214/215/216/217/218/219/220/221/222/223/224/225/226/227/228/229/230/231/232/233/234/235/236/237/238/239/240/241/242/243/244/245/246/247/248/249/250/251/252/253/254/255/256/257/258/259/260/261/262/263/264/265/266/267/268/269/270/271/272/273/274/275/276/277/278/279/280/281/282/283/284/285/286/287/288/289/290/291/292/293/294/295/296/297/298/299/300/301/302/303/304/305/306/307/308/309/310/311/312/313/314/315/316/317/318/319/320/321/322/323/324/325/326/327/328/329/330/331/332/333/334/335/336/337/338/339/340/341/342/343/344/345/346/347/348/349/350/351/352/353/354/355/356/357/358/359/360/361/362/363/364/365/366/367/368/369/370/371/372/373/374/375/376/377/378/379/380/381/382/383/384/385/386/387/388/389/390/391/392/393/394/395/396/397/398/399/400/401/402/403/404/405/406/407/408/409/410/411/412/413/414/415/416/417/418/419/420/421/422/423/424/425/426/427/428/429/430/431/432/433/434/435/436/437/438/439/440/441/442/443/444/445/446/447/448/449/450/451/452/453/454/455/456/457/458/459/460/461/462/463/464/465/466/467/468/469/470/471/472/473/474/475/476/477/478/479/480/481/482/483/484/485/486/487/488/489/490/491/492/493/494/495/496/497/498/499/500/501/502/503/504/505/506/507/508/509/510/511/512/513/514/515/516/517/518/519/520/521/522/523/524/525/526/527/528/529/530/531/532/533/534/535/536/537/538/539/540/541/542/543/544/545/546/547/548/549/550/551/552/553/554/555/556/557/558/559/560/561/562/563/564/565/566/567/568/569/570/571/572/573/574/575/576/577/578/579/580/581/582/583/584/585/586/587/588/589/590/591/592/593/594/595/596/597/598/599/600/601/602/603/604/605/606/607/608/609/610/611/612/613/614/615/616/617/618/619/620/621/622/623/624/625/626/627/628/629/630/631/632/633/634/635/636/637/638/639/640/641/642/643/644/645/646/647/648/649/650/651/652/653/654/655/656/657/658/659/660/661/662/663/664/665/666/667/668/669/670/671/672/673/674/675/676/677/678/679/680/681/682/683/684/685/686/687/688/689/690/691/692/693/694/695/696/697/698/699/700/701/702/703/704/705/706/707/708/709/710/711/712/713/714/715/716/717/718/719/720/721/722/723/724/725/726/727/728/729/730/731/732/733/734/735/736/737/738/739/740/741/742/743/744/745/746/747/748/749/750/751/752/753/754/755/756/757/758/759/760/761/762/763/764/765/766/767/768/769/770/771/772/773/774/775/776/777/778/779/780/781/782/783/784/785/786/787/788/789/790/791/792/793/794/795/796/797/798/799/800/801/802/803/804/805/806/807/808/809/810/811/812/813/814/815/816/817/818/819/820/821/822/823/824/825/826/827/828/829/830/831/832/833/834/835/836/837/838/839/840/841/842/843/844/845/846/847/848/849/850/851/852/853/854/855/856/857/858/859/860/861/862/863/864/865/866/867/868/869/870/871/872/873/874/875/876/877/878/879/880/881/882/883/884/885/886/887/888/889/890/891/892/893/894/895/896/897/898/899/900/901/902/903/904/905/906/907/908/909/910/911/912/913/914/915/916/917/918/919/920/921/922/923/924/925/926/927/928/929/930/931/932/933/934/935/936/937/938/939/940/941/942/943/944/945/946/947/948/949/950/951/952/953/954/955/956/957/958/959/960/961/962/963/964/965/966/967/968/969/970/971/972/973/974/975/976/977/978/979/980/981/982/983/984/985/986/987/988/989/990/991/992/993/994/995/996/997/998/999/1000/100

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT|<=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

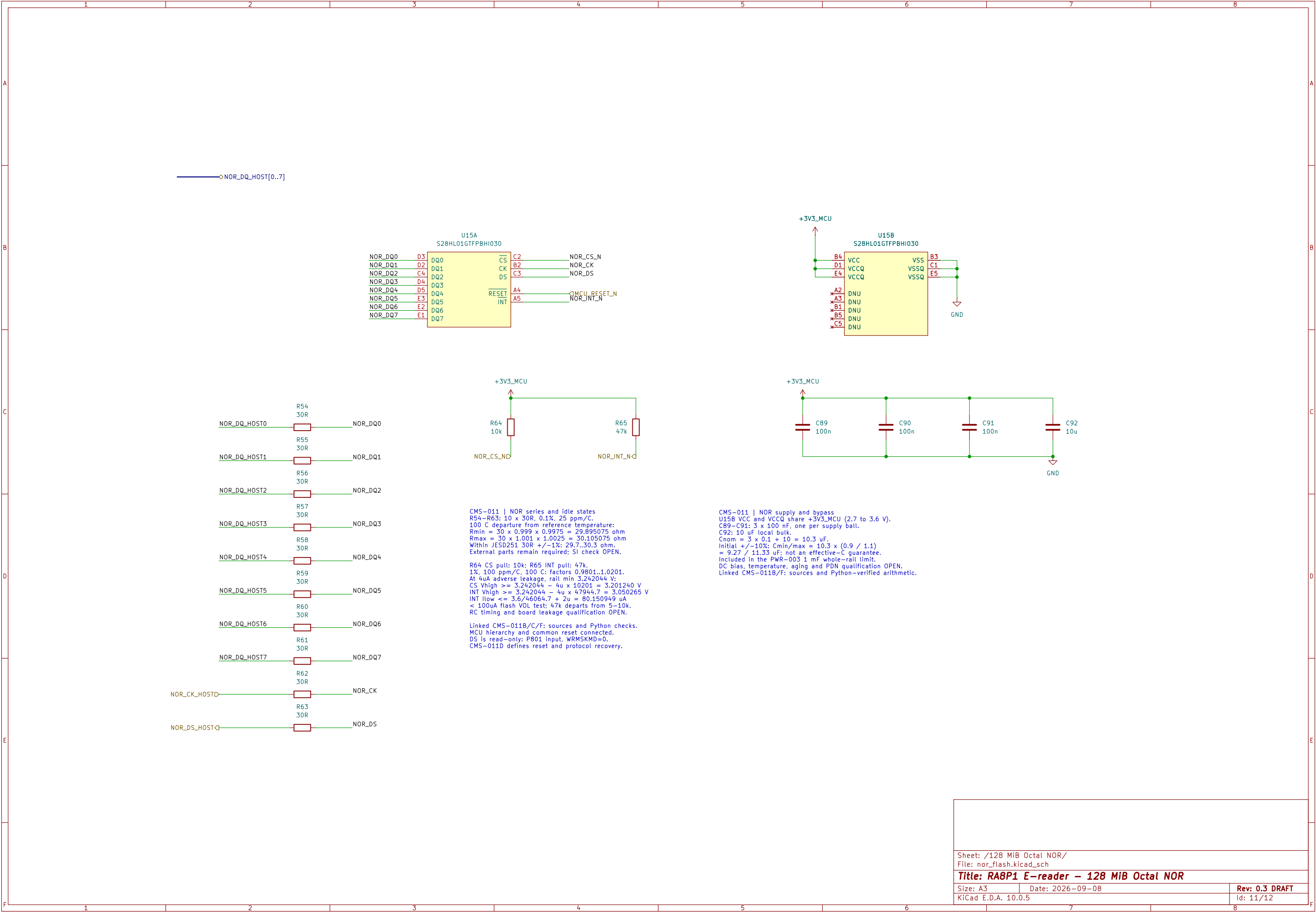
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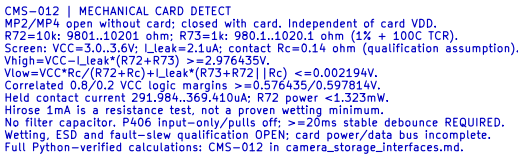
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KiCad E.D.A. 10.0.5

Rev: 0.3 DRAFT

Id: 10/12





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Size: A3	Date: 2026-09-08	Rev: 0.3 DRAFT
KiCad E.D.A. 10.0.5		Id: 12/12